

DAT NGUYEN QUOC

Senior Backend Engineer | Technical Lead

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Key Highlights

- **Scale:** Architecting systems processing **17M+ events/day** powering VNG’s top game titles
- **Performance:** Optimized workflow orchestration achieving **<3s latency SLA** at **200 req/s** throughput; load-tested the orchestration stack to **700 events/s** (~2,620 workflow starts/s)
- **Reliability:** Maintaining **99.95% uptime** across mission-critical production services
- **Leadership:** Technical Lead (since Apr 2026) managing a **7-member** cross-functional team (BE, FE, QE); drove full adoption of AI across the SDLC

Profile

Backend engineer with **7+ years** in software engineering and **5+ years** specializing in backend systems at scale. Proven track record in designing and building distributed systems handling **17M+ daily events**, leading cross-functional teams, and driving architectural decisions for mission-critical platforms. Expertise in Golang, event-driven architectures, and workflow orchestration with Temporal and Kafka. Skilled at bridging technical implementation with business requirements through direct stakeholder collaboration. Seeking Senior Backend Engineer or Technical Lead roles.

Professional Experience

Senior Software Engineer (Aug 2024) → Technical Lead (Apr 2026) <i>VNG Corporation – Game Publishing Platform</i>	Aug 2024 – Present <i>Ho Chi Minh City</i>
– Architect and develop SaaS solutions for the Game Publishing Platform, enabling automated campaign rollouts, user segmentation, real-time orchestration and multi-channel messaging for VNG’s flagship titles – Initiated and architected two core services from scratch (Segment Service, Journey Trigger), and own the event-driven pipeline from event ingestion to message delivery (see Key Projects) – Act as technical lead in stakeholder discussions with game studios, translating business requirements into technical specifications and coordinating with Product Owner and Data Studio teams – Promoted to Technical Lead (Apr 2026): scope expanded from a team of 3 engineers to a cross-functional team of 7 members (Backend, Frontend, QE), owning development roadmaps aligned with product milestones – Moved the team from siloed ownership, where each engineer knew only their own service, to shared ownership of the whole product, introducing practices that encourage engineers to work across services – Drove full adoption of AI across the SDLC: Go MCP servers, a shared engineering knowledge base and custom agents for MR review, deployment and ticket refinement Technologies: Golang, Temporal, Kafka, MongoDB, ClickHouse, Redis, Vitess, Centrifugo, OpenTelemetry, Kubernetes, ArgoCD	
Team Leader <i>Vulcan Labs – Mobile & AI Applications</i>	Jul 2023 – Aug 2024 <i>Ho Chi Minh City</i>
– Led a cross-functional team of 3 backend engineers and 3 AI engineers, taking AI features from model to production API and shipping them in the company’s mobile applications – Led the team behind AI Chat Smith, the company’s record-breaking app (see Key Projects) – Worked directly with multiple product teams, collecting requirements from Product Owners and Marketers and turning them into technical designs and delivery plans – Owned all API servers for the mobile applications, handling 1.5M+ daily requests from users worldwide – Moved deployments to GKE with monitoring (Grafana, Prometheus) and DDoS protection (Cloud Armor, Cloudflare); designed ETL pipelines for marketing metrics aggregation and visualization Technologies: Golang, GKE, Cloud Armor, Cloudflare, Redis, PostgreSQL, Grafana, Prometheus, Docker	
Senior Software Developer <i>Vulcan Labs – Mobile & AI Applications</i>	Dec 2021 – Jul 2023 <i>Ho Chi Minh City</i>
– As the company’s only backend engineer, built, shipped and maintained all backend and API services behind a fast-moving portfolio of mobile applications – Owned deployment end to end: containerized services and ran them on GCP virtual machines with Docker Compose and Docker Swarm – Established codebase architecture using Clean Architecture principles; built internal libraries for service reliability and consistency Technologies: Golang, Docker, Docker Compose, Docker Swarm, GCP Compute Engine, Redis, PostgreSQL	
Fullstack Developer <i>Pascalasia Asia Vietnam</i>	Jun 2019 – Dec 2021 <i>Ho Chi Minh City</i>

- Developed a brain data analysis platform in the R&D Department, working directly with Japanese stakeholders to translate requirements into technical solutions
 - Built the end-to-end data platform: mobile data collection, ETL pipelines, statistical processing and interactive visualization dashboards
 - Maintained and enhanced customer-facing Ruby on Rails services
- Technologies:** Python, FastAPI, VueJS, NuxtJS, Docker, Redis, Celery, MongoDB, MySQL, AWS

Key Projects

Orchestration Journey Platform

Jul 2025 – Present

VNG Corporation

Project Description:

- Event-driven automation that lets game operators react to player behaviour in real time: game events arrive on Kafka, trigger rules decide which players qualify, and Temporal workflows run multi-step journeys (delays, branches, A/B tests, external calls) ending in in-app, push, SMS or email messages
- Split into an ingestion/trigger service, a journey configuration API, a workflow worker and a reusable library of workflow node activities

Responsibility:

- Initialized the **trigger service**: Kafka consumer groups with worker-pool lifecycle management, per-source de-duplication, multiplexed event-source topics, expression-based trigger rules, frequency capping and country gating, dispatching qualified players to Temporal
- Set up the Temporal workflow worker, and extended the journey configuration API with timezone-aware time anchors for rule expressions and validation that blocks event fields colliding with reserved identifiers
- Built workflow-node features (A/B testing, cache and role-state lookup nodes, HMAC-SHA256 signed partner requests with retry classification) and made workflow context resolution panic-safe with version-gated transitions
- Optimized Temporal resource configuration to reach a **<3s latency SLA** at **200 req/s** with **99.95% uptime**
- Added end-to-end player-identity and trace context across trigger, workflow and node logs, making a single player's journey traceable in production

Tech-stack: Golang, Temporal, Kafka, expr-lang, Redis, Ristretto, MongoDB, gRPC/Gin, OpenTelemetry

Temporal Capacity & SLA Campaign

Jun 2026 – Jul 2026

VNG Corporation

Project Description:

- After Temporal persistence moved from Cassandra to Vitess (sharded MySQL), the orchestration stack needed a measured capacity and a tuned configuration before real-time campaigns could scale up

Responsibility:

- Led the capacity campaign: **45+ staging load tests** with Grafana/Prometheus monitoring, attributing each bottleneck to a single stack layer before tuning it
- Tuned Temporal history, matching and frontend services (shard IO concurrency, memory, sticky cache, pod spreading across nodes) through GitOps changes with reversible rollouts
- Raised the load-tested clean ceiling **1.75x, from 400 to 700 events/s** (**~2,620 workflow starts/s**) with every Temporal layer at p95 under 400ms and database p95 ~4ms
- Identified the next limit: trigger ingestion is latency-bound by synchronous per-message gate checks, not CPU, which set the roadmap for the next optimization

Tech-stack: Temporal, Vitess, Kubernetes, ArgoCD, Helm, k6, Grafana, Prometheus

Segment Service

Dec 2024 – Present

VNG Corporation

Project Description:

- Builds player audiences (past behaviour, churn, RFM and imported segments) from the data warehouse so game operators can target campaigns without writing SQL
- Segments are versioned and can be refreshed on demand or in the background against current customer data

Responsibility:

- Initialized the service from the team's Go service template
- Engineered a **dynamic query builder** over ClickHouse fact and CDP tables: product and lifecycle filters, purchase and attribution-event conditions, and dry-run previews before a segment is saved
- Built segment import from files (Parquet on S3) and a dimension-mapping pipeline that standardizes phone numbers and enriches players with device tokens for every delivery channel
- Implemented Temporal workflows for segment refresh and versioning, with idempotent on-demand refresh, heartbeat handling and trace propagation for background jobs

Tech-stack: Golang, ClickHouse, MongoDB, Temporal, S3/Parquet, gRPC, OpenTelemetry

Multi-channel Messaging & Delivery

Jan 2025 – Present

VNG Corporation

Project Description:

- Delivers campaign messages to players over in-app messaging, push, SMS and webview, and lets external partners send notifications through scoped APIs

Responsibility:

- Built partner integrations: tenant-scoped Kafka topics, API keys scoped by game and product with rate-limit quotas and idempotency, and a pluggable partner signing/encryption layer with key versioning
- Added delivery-funnel and ACK metrics per game and campaign, and surfaced real-time (Centrifugo) publish rejections as delivery errors with fail-closed handling
- Instrumented SDK event ingestion with per-campaign Prometheus metrics, with cardinality agreed with SRE

Tech-stack: Golang, Kafka, Centrifugo, MongoDB, Redis, Temporal, Prometheus, OpenTelemetry

Campaign Portal Backend & Shared Go Platform Library

Aug 2024 – Present

VNG Corporation

Project Description:

- The portal backend is the entry point for operators, managing campaigns and segments and routing to downstream services; the shared Go library provides HTTP, logging, data clients, Kafka, Temporal and telemetry to every team service

Responsibility:

- Built multi-market routing in the portal (fan-out across market-scoped domains with validation) behind SSO and role-based access control
- Maintain the shared library: OpenTelemetry tracing and Temporal trace propagation, self-healing Kafka consumers, HMAC request-signing helpers with key rotation, and log redaction of sensitive fields by default
- Manage service deployments through GitOps (Helm values, ArgoCD) and keep dependencies patched against HIGH/CRITICAL CVEs

Tech-stack: Golang, Gin, gRPC, MongoDB, Kafka, Temporal, OpenTelemetry, Helm, ArgoCD, GitLab CI

AI-Assisted SDLC Tooling

Apr 2026 – Present

VNG Corporation

Project Description:

- Internal tooling that gives AI coding agents safe access to platform knowledge, debugging data and test/staging APIs, adopted across the team’s development workflow

Responsibility:

- Built Go **MCP servers** for platform debugging (logs, metrics, traces) and authenticated test/staging API calls
- Built a platform knowledge base with semantic search, kept in sync with the source code, plus a CLI that installs and configures it for every engineer
- Created custom agents and skills for MR review, deployment, Jira ticket refinement and sprint sync

Tech-stack: Golang, MCP, Claude Code, Cobra, Bubble Tea

AI Ecosystem Platform & AI Chat Smith

2022 – 2024

Vulcan Labs

Project Description:

- Cross-platform ecosystem consolidating Vulcan’s AI products under unified accounts and subscriptions
- AI Chat Smith, a mobile ChatGPT app for global users, launched as an MVP and broke the company’s previous records, smoothly serving **200–300 requests/s**

Responsibility:

- Architected the ecosystem: user management and subscription systems supporting multi-platform access
- Implemented credit-based monetization with fraud prevention and automated renewal handling
- Built the ChatGPT service handling **1.5M+ daily requests**: cost-optimized API layer with abuse prevention and robust error handling, deployed on GKE with Cloud Armor and a full observability stack

Tech-stack: Golang, PostgreSQL, Redis, GKE, Cloud Armor, Cloudflare, Grafana, Prometheus

Technical Skills

Languages	Golang (Expert), Python, SQL
Backend & APIs	gRPC, grpc-gateway, REST (Gin), Event-Driven Microservices, Rule Engines
Workflow & Streaming	Temporal, Kafka, Centrifugo
Data	MongoDB, ClickHouse, Redis, Vitess (MySQL), PostgreSQL, Elasticsearch, BigQuery
Observability	OpenTelemetry, Prometheus, Grafana
Platform & Delivery	Kubernetes, Helm, ArgoCD (GitOps), GitLab CI, Docker, GCP, AWS
AI Engineering	Claude Code, MCP Servers (Go), AI Agent Workflows, AI-Assisted SDLC
Leadership	Team of 7 (BE/FE/QE), Roadmapping, Stakeholder Management, Process Improvement

Education

Bachelor of Engineering – Information Security

Nov 2014 – Mar 2019

Posts and Telecommunications Institute of Technology

Ho Chi Minh City

Languages

English (Professional working proficiency) | Vietnamese (Native)